Exercise §2.4.17 Solution

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Subgroups

1.2 Appendix: Partially Ordered Sets and Zorn's Lemma

Definition 0.0.1 (poset)**.** A *partial order* on a nonempty set A is a relation $\leq$ on A satisfying

1. $x \leq x$ for all $x \in A$ (reflexive),
2. if $x \leq y$ and $y \leq z$ then $x = y$ for all $x, y \in A$ (antisymmetric),
3. if $x \leq y$ and $y \leq z$ then $x \leq z$ for all $x, y, z \in A$ (transitive).

We usually say that A is a partially ordered set under the ordering $\leq$ or that A is partially ordered by $\leq$.

Definition 0.0.2. Let the nonempty set A be partially ordered by $\leq$.

1. A subset $B \subset A$ is called a *chain* if for all $x, y \in B$, either $x \leq y$ or $y \leq x$.
2. An *upper bound* for a subset $B \subset A$ is an element $u \in A$ such that $b \leq u$, for all $b \in B$.
3. A *maximal element* of A is an element $m \in A$ such that if $m \leq x$ for any $x \in A$, then $m = x$.

In the literature a chain is also called a *tower* or called a *totally ordered* or *linearly ordered* or *simply ordered* subset.

Lemma 1 (Zorn's Lemma)**.** *If A is a nonempty partially ordered set in which every chain has an upper bound then A has a maximal element.*

2.4 Subgroups Generated by Subsets of a Group

§2.4: 17. This is an exercise involving Zorn's Lemma (see Appendix I) to prove that every nontrivial finitely generated group possesses maximal subgroups. Let G be a group, say $G = \langle g_1, g_2, \ldots, g_n \rangle$ and let $\mathcal{S}$ be the set of all proper subgroups of G. Then $\mathcal{S}$ is partially ordered by inclusion.

Let $\mathcal{C}$ be a chain in $\mathcal{S}$.

(a) Prove that the union, H, of all the subgroups in $\mathcal{C}$ is a subgroup of G.

(b) Prove that H is a *proper* subgroup. [If not, each g_i must lie in H and so must lie in some element of the chain $\mathcal{C}$. Use the definition of a chain to arrive at a contradiction.]

(c) Use Zorn's Lemma to show that $\mathcal{S}$ has a maximal element (which is, by definition, a maximal subgroup.)

Proof. (Hernandez)

Let $\mathcal{C}$ be a chain in the set $\mathcal{S}$ of all proper subgroups partially ordered by inclusion of a group $G = \langle g_i \rangle_{i=1}^{n}$ generated by a finite number $n \in \mathbb{N}$ of generators.

Consider two elements of the union of all the subgroups in $\mathcal{C}$:

$$x, y \in H = \bigcup_{H_\alpha \in \mathcal{C}} H_\alpha. \tag{1}$$

Then, there exist subgroups $H_1, H_2 \in \mathcal{C}$ such that $x \in H_1$ and $y \in H_2$. By the definition of a chain, we assume without loss of generality that $H_1 \leq H_2$.

(a) Then H is a group because it contains the following: $xy, e, x^{-1} \in \langle x, y \rangle \subset H_2 \subset H$.

(b) Suppose that H is not a proper subgroup, namely that $H = G$. Since $g_i \in H$, there exist subgroups $H_i \in \mathcal{C}$ such that $g_i \in H_i$. By the definition of chain, there exists a permutation $\sigma \in S_n$ such that $H_{\sigma(i)} \leq H_{\sigma(j)}$ for $1 \leq i \leq j \leq n$. Then, each $g_i \in H_{\sigma(n)}$, so any $g \in G$ is in $H_{\sigma(n)}$, which contradicts its propriety. Therefore, H is a proper subgroup.

(c) According to the previous parts, $H_\alpha \in H$ for all $H_\alpha \in \mathcal{C}$. By definition, H is an upper bound for $\mathcal{C} \subset \mathcal{S}$. By Zorn's Lemma, $\mathcal{S}$ has a maximal element. Therefore, G has a maximal subgroup.

This concludes the proof of the following theorem. □

Theorem. *There exists a maximal proper subgroup for any finitely generated group.*

Question. *When does a finitely generated group have a* unique *maximal subgroup?*